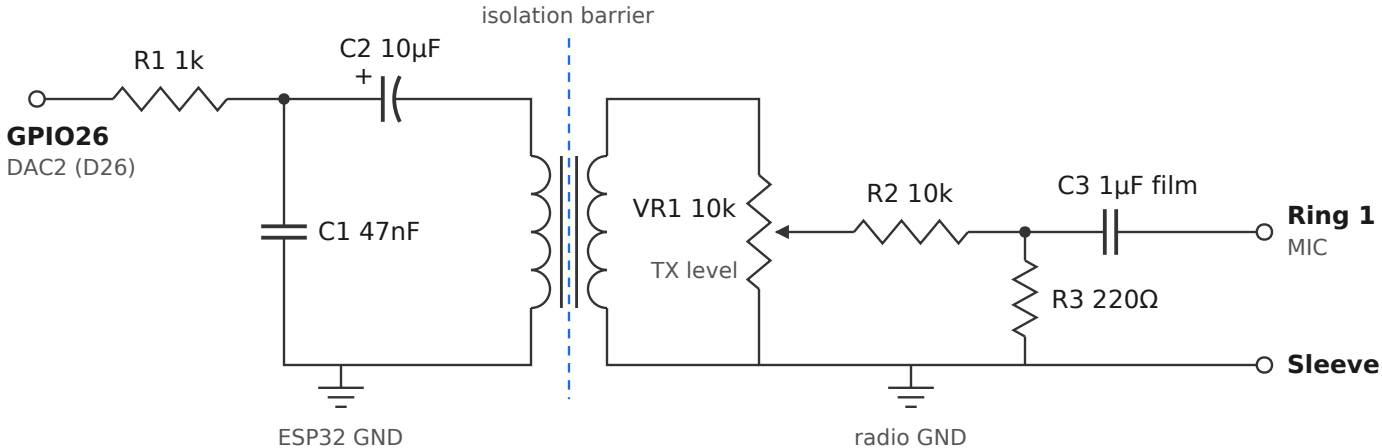
TX audio

ESP32 NodeMCU (30-pin, ESP32-WROOM-32) to QYT KT-8900D rear 3.5mm TRRS "Speaker" jack

TX: DAC tone is filtered, isolated, then padded down to mic level



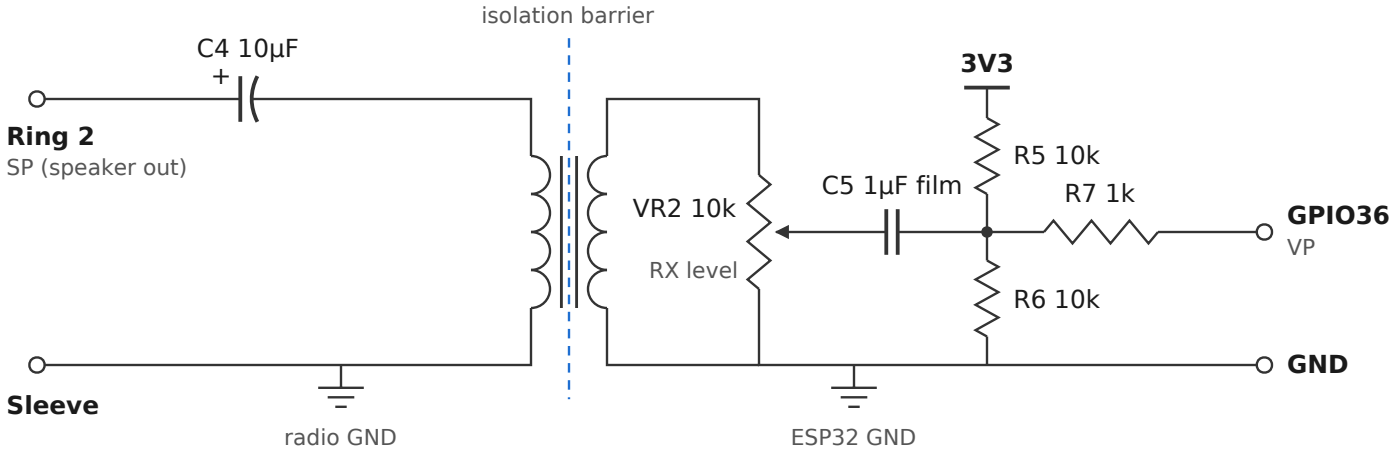
T1 Bourns LM-NP-1001-B1L, 600:600

- Firmware (ESP32APRS_Audio) outputs TX audio on DAC2 = GPIO26, labeled D26 on the board.
- Left of the barrier uses ESP32 GND; right of it uses radio GND (TRRS Sleeve). Only T1 crosses.
- VR1 ends go across T1's secondary; the wiper feeds the R2/R3 pad (~46:1). Start low, raise for ~3 kHz deviation.
- C1 47nF with R1 1k gives a ~3.4 kHz low-pass. C3 blocks the mic jack's electret bias.

RX audio

ESP32 NodeMCU (30-pin, ESP32-WROOM-32) to QYT KT-8900D rear 3.5mm TRRS "Speaker" jack

RX: speaker audio is isolated, trimmed, and centered at 1.65 V



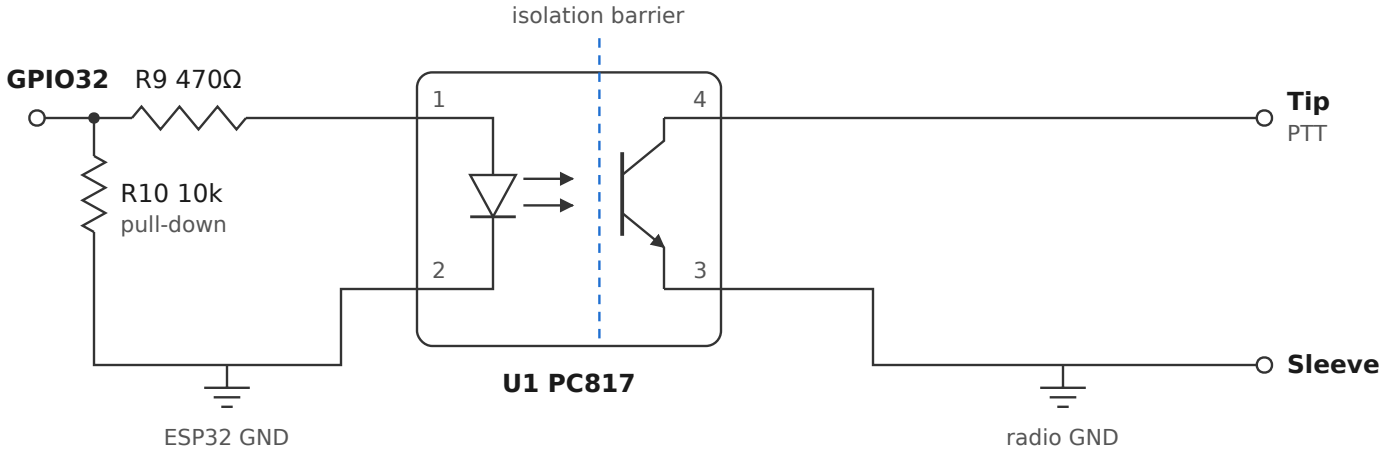
T2 Bourns LM-NP-1001-B1L, 600:600

- Radio on the left, ESP32 on the right; only T2 crosses the isolation barrier.
- R5/R6 hold GPIO36 (VP) at 1.65 V; C5 couples the trimmed audio onto that bias point.
- Set ADC Attenuation to 11dB in the MOD tab so 1.65 V sits inside the ADC range.
- Adjust VR2 so strong stations read about -10 to -15 dBV in Last Heard.

PTT

ESP32 NodeMCU (30-pin, ESP32-WROOM-32) to QYT KT-8900D rear 3.5mm TRRS "Speaker" jack

PTT: an optocoupler keys the radio with no shared ground



- GPIO32 high lights the LED (pins 1-2) and the phototransistor pulls Tip to Sleeve (pins 4-3), keying the radio.
- Set PTT GPIO 32, Active HIGH in the MOD tab.
- R10 holds GPIO32 low during boot so the radio can't key at power-up.
- Pin numbers are for the DIP-4 PC817 package.